



AI-BASED PREDICTIVE MAINTENANCE

Smart Machine Health & Early Warning System

Predicting machine failure risk from operating data — built on the UCI AI4I 2020 dataset

Decision Tree

Random Forest

Health Score

THE PROBLEM

Machine Failures Are Costly — and Sudden

Machines can stop working because of high temperature, high torque, unusual speed, or worn tools. Repairing a machine after it breaks costs money and halts production.



Unexpected Failure

A machine breaks down with no warning.



Downtime & Repair Cost

Production stops; repairs are expensive.



Need for Early Warning

Predicting risk earlier avoids the surprise.

THE CONCEPT

What Is Predictive Maintenance?

Using data to predict a possible machine problem — before the machine actually fails.

The AI model studies thousands of past examples and learns the patterns that link machine conditions to failures. Give it new readings, and it predicts the risk.

EXAMPLE MACHINE READING

Air Temperature	300 K
Process Temperature	310 K
Rotational Speed	1,500 rpm
Torque	40 Nm
Tool Wear	150 minutes

THE DATASET

UCI AI4I 2020 Predictive Maintenance Dataset

A free, simulated dataset — large enough to learn from, small enough to run on a normal computer.

10,000

machine records



Air Temperature



Process Temperature



Rotational Speed



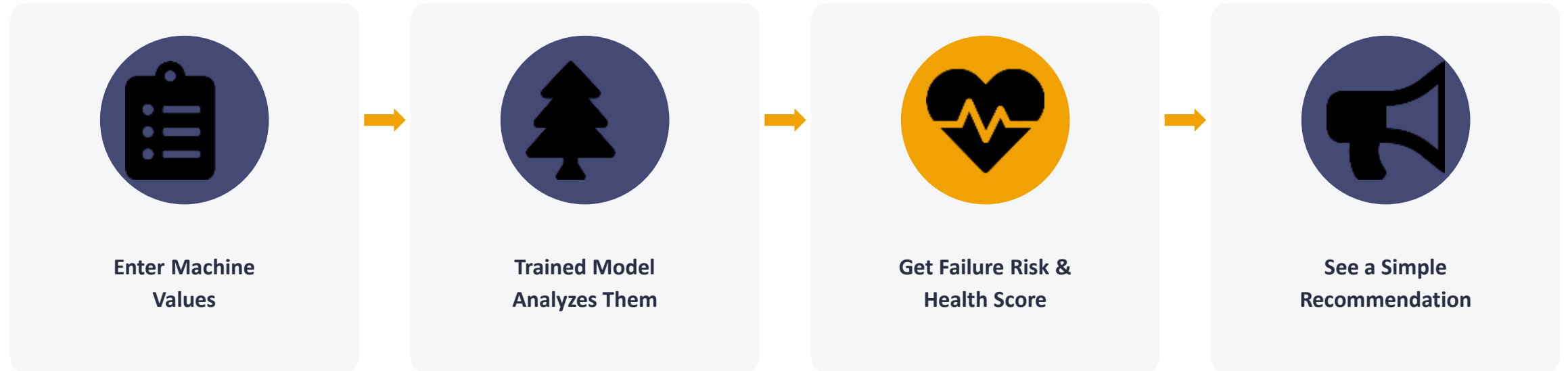
Torque



Tool Wear

+ a Failure column marking whether the machine actually failed — this is the target the model learns to predict.

How the System Works



Decision Tree → Random Forest



STEP 1

Decision Tree

Easy to understand — it makes decisions by asking questions like “Is tool wear above X?” A great starting baseline.



STEP 2 — STRONGER BASELINE

Random Forest

Combines many decision trees and averages their answers — usually more accurate and more reliable than any single tree.

EXAMPLE RESULT

What the User Sees

STATUS: WARNING

72%

FAILURE RISK

28/100

HEALTH SCORE

MAIN RISK FACTOR

High Tool Wear

RECOMMENDATION

Inspect the machine and consider maintenance soon.

Everything Used Is Free



Python

The programming language for the AI part.



Pandas

A tool for loading and working with data in table form



NumPy

A tool for doing fast math on large sets of numbers.



Scikit-learn

Train and test the machine-learning model.



Matplotlib

Simple graphs to explore the data.



Streamlit

Optional simple web interface for the demo.

Project Access & Live Demonstration



A

Git Repository

Access the complete source code, machine learning pipeline, data pre-processing scripts, and architecture documentation.

 github.com/pawankumar0new/predictive-maintenance-ai



B

Deployed Application

Interact directly with the live Streamlit dashboard to test input parameters, evaluate real-time telemetry, and view model predictions.

 predictive-maintenance-ai-cg7awmxo9yyshto5fzudof.streamlit.app



Conclusion

Smart Machine Health & Early Warning System analyzes machine operating conditions — temperature, speed, torque, and tool wear — to predict failure risk, and presents it as a simple health score, risk level, and maintenance recommendation.

Predict

Score

Explain

Recommend

Thank you